

11-3

可见光三波长为 $0.38 \sim 0.76 \mu\text{m}$

$$\lambda_1 T = 0.38 (273.15 + 2000) = 863.797 \mu\text{m} \cdot \text{K}$$

$$\lambda_2 T = 0.76 (273.15 + 2000) = 1727.594 \mu\text{m} \cdot \text{K}$$

$$F_b(\lambda_1 T - \lambda_2 T) = F_b(0 - \lambda_2 T) - F_b(0 - \lambda_1 T) = 0.032268 - 0 = 0.032268 \\ = 3.23 \%$$

11-6

$$\varepsilon = \frac{E}{E_b} = \frac{\int_0^{\infty} \varepsilon_{\lambda} E_{b\lambda} d\lambda}{E_b} = \frac{\int_0^{\lambda_1} \varepsilon_{\lambda} E_{b\lambda} d\lambda}{E_b} + \frac{\int_0^{\lambda_2} E_{b\lambda} \varepsilon_{\lambda} d\lambda}{E_b} + \frac{\int_0^{\lambda_3} \varepsilon_{\lambda} E_{b\lambda} d\lambda}{E_b}$$

$$\varepsilon = \varepsilon_{\lambda_1} F_{b(0-\lambda_1 T)} + \varepsilon_{\lambda_2} (F_{b(0-\lambda_2 T)} - F_{b(0-\lambda_1 T)}) + \varepsilon_{\lambda_3} (F_{b(0-\lambda_3 T)} - F_{b(0-\lambda_2 T)})$$

$$T = 1473.15 \text{ K}$$

$$\varepsilon_1 = 0.1 \quad \varepsilon_2 = 0.4 \quad \varepsilon_3 = 0.2$$

$$\lambda_1 T = 1473.15 \mu\text{m} \cdot \text{K}$$

$$F_{b(0-\lambda_1 T)} = 0.01215 = 1.215 \%$$

$$\lambda_2 T = 4419.45 \mu\text{m} \cdot \text{K}$$

$$F_{b(0-\lambda_2 T)} = 0.5518 = 55.18 \%$$

$$\lambda_3 T = 7365.75 \mu\text{m} \cdot \text{K}$$

$$F_{b(0-\lambda_3 T)} = 0.8278 = 82.78 \%$$

$$\varepsilon = 0.272275$$

$$E = \varepsilon E_b = 0.272275 (5.67 \times 10^{-8}) (1473.15)^4 \\ = 72.7074 \text{ kW/m}^2$$

11-8

$$E_b = \sigma_b T^4 = 217818.72 \text{ W/m}^2$$

$$\lambda_{\text{max}} = \frac{\lambda_{\text{max}}}{T} \text{ (Wien 位移定律)} = \frac{2.8976 \times 10^{-3}}{1400} = 2.0697 \mu\text{m}$$